

DIS Department of Mathematics
MYP Grade 10 Extended – Assessment on rules for radicals

Name: Answers Date: _____

You will be assessed against criterion C. You may not use your calculator for any question. Show all your working.

1. Simplify the following expressions as far as possible!

(a) $5\sqrt{18} - 2\sqrt{2} + 3\sqrt{8}$

$$\begin{aligned} &= 5\sqrt{9 \cdot 2} - 2\sqrt{2} + 3\sqrt{4 \cdot 2} \\ &= 13\sqrt{2} + 6\sqrt{2} \\ &= \underline{\underline{19\sqrt{2}}} \end{aligned}$$

(2 marks)

(b) $8 - \sqrt{3} + 9\sqrt{3} - \sqrt{27} + 3$

$$\begin{aligned} &= 11 + 8\sqrt{3} - \sqrt{9 \cdot 3} \\ &= \underline{\underline{11 + 5\sqrt{3}}} \end{aligned}$$

(2 marks)

(Total 4 marks)

2. (a) The decimal value of $\sqrt{12}$ is between the two consecutive integers 3 and 4.

(1 mark)

(b) Show why $\sqrt{\frac{10}{3}} = \frac{\sqrt{30}}{3}$.

$$\sqrt{\frac{10}{3}} = \frac{\sqrt{10}}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{30}}{3} \quad (\text{shown})$$

(1 mark)

(c) Show why $\sqrt{28} = 2\sqrt{7}$.

$$\sqrt{28} = \sqrt{7 \cdot 4} = \sqrt{4} \sqrt{7} = \underline{\underline{2\sqrt{7}}}$$

(shown)

(1 mark)

(d) Expand and fully simplify $5(2 + \sqrt{3})(1 - \sqrt{3})$.

$$= 5(2 - 2\sqrt{3} + \sqrt{3} - 3)$$

$$= 5(-1 - \sqrt{3})$$

$$= \underline{\underline{-5 - 5\sqrt{3}}}$$

(2 marks)

(e) Expand and fully simplify $(6 + \sqrt{5})(1 - \sqrt{5}) - \sqrt{8} + 2\sqrt{20} - \sqrt{5} + 3\sqrt{2} - 1$.

$$= 6 - 6\sqrt{5} + \sqrt{5} - 5 - \sqrt{8} + 4\sqrt{5} - \sqrt{5} + 3\sqrt{2} - 1$$

$$= \underline{\underline{-2\sqrt{5} + \sqrt{2}}}$$

(2 marks)

(f) Expand and fully simplify $(1 + \sqrt{2})^4$.

$$(1 + \sqrt{2})^2 (1 + \sqrt{2})^2$$

$$= (1 + 2\sqrt{2} + 2)(1 + 2\sqrt{2} + 2)$$

$$= (3 + 2\sqrt{2})^2$$

$$= 9 + 12\sqrt{2} + 8$$

$$= \underline{\underline{17 + 12\sqrt{2}}}$$

(2 marks)

(Total 9 marks)

3. Express the following with a **positive** integer denominator. Simplify as far as possible.

$$(a) \frac{4}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{4\sqrt{3}}{3}$$

(1 mark)

$$(b) \frac{5}{\sqrt{3}-1} \cdot \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{5\sqrt{3}+5}{3-1} = \frac{5\sqrt{3}+5}{2}$$

(2 marks)

$$\begin{aligned} (c) \frac{4-\sqrt{2}}{\sqrt{2}+4} \cdot \frac{\sqrt{2}-4}{\sqrt{2}-4} &= \frac{(4-\sqrt{2})(\sqrt{2}-4)}{2-16} \\ &= \frac{4\sqrt{2}-16-2+4\sqrt{2}}{-14} = \frac{8\sqrt{2}-18}{-14} \\ &= \frac{9-4\sqrt{2}}{7} \end{aligned}$$

(2 marks)

$$\begin{aligned} (d) \frac{\sqrt{3}+1}{\sqrt{3}-3\sqrt{2}} \cdot \frac{\sqrt{3}+3\sqrt{2}}{\sqrt{3}+3\sqrt{2}} &= \frac{(\sqrt{3}+1)(\sqrt{3}+3\sqrt{2})}{3-18} \\ &= \frac{3+3\sqrt{6}+\sqrt{3}+3\sqrt{2}}{-15} \\ &= \frac{-3-3\sqrt{2}-\sqrt{3}-3\sqrt{6}}{15} \end{aligned}$$

(2 marks)

(Total 7 marks)

4. The following expression can be written in the form $a + b\sqrt{5}$, where $a, b \in \mathbb{Q}$. Find the values of a and b .

$$\begin{aligned}
 \frac{2\sqrt{5}}{4+\sqrt{5}} - \frac{\sqrt{5}}{4-\sqrt{5}} &= \frac{2\sqrt{5}(4-\sqrt{5})}{(4+\sqrt{5})(4-\sqrt{5})} - \frac{\sqrt{5}(4+\sqrt{5})}{(4+\sqrt{5})(4-\sqrt{5})} \\
 &= \frac{8\sqrt{5} - 10 - 4\sqrt{5} - 5}{(4+\sqrt{5})(4-\sqrt{5})} = \frac{4\sqrt{5} - 15}{4 \cdot 4 - 5} \\
 &= \frac{4\sqrt{5} - 15}{11} = -\frac{15}{11} + \frac{4}{11}\sqrt{5} \\
 \therefore a &= -\frac{15}{11}, b = \frac{4}{11}
 \end{aligned}$$

(Total 3 marks)

5. Write $\sqrt{\frac{2+\sqrt{3}}{2-\sqrt{3}}}$ in the form $a + \sqrt{a+1}$, where $a \in \mathbb{N}$.

$$\begin{aligned}
 &= \sqrt{\frac{2+\sqrt{3}}{2-\sqrt{3}} \cdot \frac{2+\sqrt{3}}{2+\sqrt{3}}} = \sqrt{\frac{4+4\sqrt{3}+3}{1}} \\
 &= \sqrt{4\sqrt{3}+7} = \sqrt{(2+\sqrt{3})^2} \\
 &= \underline{\underline{2+\sqrt{3}}}
 \end{aligned}$$

(Total 3 marks)

6. Solve the linear equation below. Write your solution in simplest form with no radicals or negatives in the denominator.

$$2\sqrt{2}x - \sqrt{5} = 3\sqrt{5} - x$$

Note: The "x" on the left hand side of the equation is **not** underneath the square root.

$$2\sqrt{2}x + x = 4\sqrt{5}$$

$$x(2\sqrt{2} + 1) = 4\sqrt{5}$$

$$x = \frac{4\sqrt{5}}{2\sqrt{2} + 1}$$

$$x = \frac{4\sqrt{5}}{2\sqrt{2} + 1} \cdot \frac{2\sqrt{2} - 1}{2\sqrt{2} - 1} = \frac{8\sqrt{10} - 4\sqrt{5}}{7}$$

(Total 2 marks)

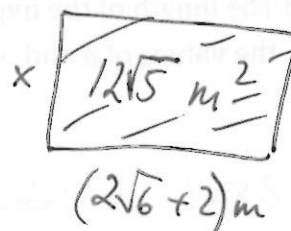
7. A rectangle has area $12\sqrt{5} \text{ m}^2$ and one of the rectangle's side lengths is $(2\sqrt{6} + 2) \text{ m}$. Find the length of the other side of the rectangle. Give your answer in the form $\frac{a\sqrt{30} - a\sqrt{5}}{5} \text{ m}$, where $a \in \mathbb{N}$.

$$12\sqrt{5} = (2\sqrt{6} + 2)x$$

$$x = \frac{12\sqrt{5}}{2\sqrt{6} + 2} = \frac{6\sqrt{5}}{\sqrt{6} + 1}$$

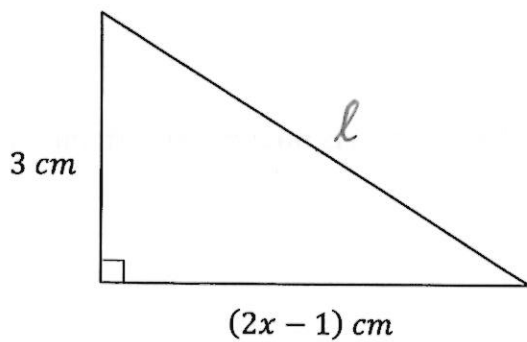
$$= \frac{6\sqrt{5}(\sqrt{6} - 1)}{(\sqrt{6} + 1)(\sqrt{6} - 1)} = \frac{6\sqrt{30} - 6\sqrt{5}}{6 - 1}$$

$$= \frac{6\sqrt{30} - 6\sqrt{5}}{5}$$



(Total 3 marks)

8. The diagram below shows a right triangle with side lengths 3 cm and $(2x - 1)$ cm.



- (a) Given that the triangle's area is $(1 + \sqrt{3})$ cm², show that $x = \frac{5 + 2\sqrt{3}}{6}$ cm.

$$\frac{3}{2}(2x - 1) = 1 + \sqrt{3}$$

$$3(2x - 1) = 2 + 2\sqrt{3}$$

$$6x - 3 = 2 + 2\sqrt{3} \rightarrow 6x = 5 + 2\sqrt{3} \rightarrow x = \frac{5 + 2\sqrt{3}}{6}$$

(shown)

(2 marks)

- (b) The length of the hypotenuse can be expressed as $\frac{\sqrt{a + b\sqrt{3}}}{3}$ cm, where $a, b \in \mathbb{N}$. Find the values of a and b .

$$2x - 1 \rightarrow \text{when } x = \frac{5 + 2\sqrt{3}}{6}, \quad 2x - 1 = \frac{5 + 2\sqrt{3}}{3} - \frac{3}{3} \\ = \frac{2 + 2\sqrt{3}}{3}$$

$$3^2 + \left(\frac{2 + 2\sqrt{3}}{3}\right)^2 = l^2$$

$$\rightarrow l = \sqrt{9 + \frac{4 + 8\sqrt{3} + 12}{9}} = \sqrt{\frac{81}{9} + \frac{16 + 8\sqrt{3}}{9}} = \sqrt{\frac{97 + 8\sqrt{3}}{9}} \\ = \frac{\sqrt{97 + 8\sqrt{3}}}{3}$$

$$\therefore \underline{\underline{a = 97, b = 8}}$$

(2 marks)

(Total 4 marks)